Meridian

A state-space neural forecaster of daily financial volatility: audited benchmark, tail-risk evaluation and forward test

Project status report

Report date: November 3, 2026

Scope of this report

This report covers the complete project in the `ProjectWM` repository: the original pre-registered protocol, the history of reported results, the audit that withdrew the earlier numbers, the re-run benchmarks, the frozen forward test, and the remaining work. **Every result number is generated programmatically** from the saved prediction and results files (`scripts/build_report_tex.py`); none is typed by hand. Numbers produced before the audit are reported only as history and are explicitly marked as withdrawn.

Repository: `ProjectWM`, branch `harness-audit` at commit `0a6cfcc` (5 commits ahead of `main` = `ab041b0`, not yet pushed)

Abstract

Meridian forecasts next-day realized variance of financial assets with a diagonal state-space (SSM) encoder, a joint-embedding predictive (JEPA) auxiliary objective and a head trained directly on the QLIKE loss, initialized at the HAR-RV solution. We evaluate it on 11 assets (8 US equities/ETFs and 3 FX pairs, daily data 2007-01-01 to 2026-07-31) with a purged, embargoed walk-forward protocol. Model choices were made only on a development period (2012-06-14 to 2018-12-31); results are reported on an untouched holdout (2019-01-01 to 2026-06-29, 2,168 trading dates, 20,906 forecasts per model). An audit of the earlier evaluation code found eight defects, including significance tests computed on correlated panel rows and 36 stale plus 469 malformed FX price bars; all earlier numbers were withdrawn and the benchmarks re-run.

On the holdout, calibrated Meridian reduces QLIKE loss by -7.5% [-11.3, -4.7] relative to per-asset HAR-RV (Diebold–Mariano p <0.001), beats HAR on 11/11 assets and is the best of 15 forecasters; the strongest classical/ML alternative, LightGBM, achieves -5.3%. Against the strongest HAR form (raw per-asset HAR) the reduction is -6.8% [-9.7, -4.5]. The pre-registered bar ($\geq 5\%$ reduction with $p < 0.05$) is met by the point estimate, but the confidence interval extends to -4.5%, so a gain above 5% is likely rather than certain. Three proposed architectural improvements did not help. For next-day Value-at-Risk and Expected Shortfall, a Meridian-filtered EVT model is the best of 13 risk models (FZ0 loss difference -0.1218 vs GJR-GARCH with filtered historical simulation, p <0.001), but its advantage over a HAR-filtered tail model is not statistically significant. A frozen forward test started on 2026-11-01 16:19:03 UTC; its first scored day is unfavourable to Meridian (QLIKE $+13.0\%$ vs HAR on 11 forecasts), which is uninformative at this sample size. We estimate that roughly 451 trading days of forward data are needed to confirm the holdout effect with 80% power.

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1 Introduction and objectives

Daily volatility forecasts drive risk limits, position sizing and option pricing. The heterogeneous autoregressive model of realized variance (HAR-RV) [8] is the standard benchmark: it is simple, robust and notoriously hard to beat out of sample. The goal of the Meridian project is to test, under a protocol fixed in advance, whether a modern sequence model can beat HAR-RV and the other standard volatility models by a material and statistically significant margin, and whether that advantage carries over to tail-risk forecasts (Value-at-Risk and Expected Shortfall).

Pre-registered objective. The evaluation protocol was locked on 2026-07-29 (`PREREGISTRATION.md`) before the model was trained. Its primary win criterion is:

Primary win criterion (PREREGISTRATION.md §5)

Meridian beats HAR-RV if, on purged out-of-sample data pooled across the universe, it achieves **at least a 5% relative reduction in QLIKE loss** and a **one-sided Diebold–Mariano test rejects equal accuracy at** $p < 0.05$ (HAC variance).

A secondary criterion concerned regime detection versus a hidden Markov model; it has not been re-evaluated under the audited harness (section 12).

What this report is. A complete, end-to-end account of the work in the repository: the model, the data, the protocol, the history of results (including the ones that were wrong), the current validated results, the forward test, and the remaining work with time estimates. Section 7 explains why every result reported before 2026-10-31 has been withdrawn.

2 Project timeline and status at a glance

Table 1: Project timeline. Dates are taken from `PREREGISTRATION.md` and the git history.

Date	Event
2026-07-29	Evaluation protocol and win criteria locked (`PREREGISTRATION.md`). Amendment 1 (single pre-committed 5-seed ensemble) and Amendment 2 (regime evaluation redesign).
2026-07-30	Amendment 3: evaluation plan for the bridged world model (volatility, regime, tail).
2026-07-31	Amendment 4: the benchmark script evaluates the real neural core (it had been a linear stand-in). Amendment 5: HAR-anchored initialization, walk-forward calibration, forecast combination.
2026-09-08	Commit `ab041b0` pushed to GitHub (`main`), claiming a QLIKE reduction vs HAR (later withdrawn).
2026-10-31	Amendment 6: audit of the benchmark harness; eight defects; all earlier numbers withdrawn; dev/holdout selection rule declared before any re-run result was seen.
2026-11-01	Re-run volatility benchmark (15 models); documentation cleaned of withdrawn numbers; Amendment 7: tail-risk benchmark (13 models) and forward test declared, then run; models frozen.
2026-11-03	First forward-test outcomes scored (1 day).

3 Data

Daily open/high/low/close bars come from the Yahoo Finance chart API for 8 US equities and ETFs (SPY, QQQ, IWM, AAPL, MSFT, JPM, XOM, JNJ) and 3 FX pairs (EURUSD, USDJPY, GBPUSD), plus two out-of-universe assets used only as a generalization check (TLT, GLD). VIX and the 10- and 2-year Treasury yields come from FRED (2007-01-02 to 2026-07-28). All data are free and delayed.

Table 2: Status at a glance (validated numbers only).

Question	Current answer	Status
Does Meridian beat per-asset HAR-RV on 1-day volatility (holdout)?	-7.5% [-11.3, -4.7], p <0.001; 11/11 assets	Yes
Against the strongest HAR form?	-6.8% [-9.7, -4.5], p <0.001	Yes
Is the gain reliably $\geq 5\%$?	95% interval reaches -4.5%	Uncertain
Best of 15 volatility models?	Best single model; combination -8.2%	Yes
Do proposed improvements M1–M3 help?	No (development period)	No
Better tail risk (VaR/ES) than GARCH-family models?	FZ0 Δ -0.1218, p <0.001	Yes
Better tail risk than HAR-based tails?	not significant (p 0.076 at 1%, 0.369 at 2.5%)	Not shown
Forward (live) confirmation?	1 day scored; needs ≈451 days	Pending
Regime module vs HMM?	not re-evaluated after the audit	Open
Trading profitability (alpha)?	not claimed	—

Target. Daily variance is measured with the Garman–Klass range estimator [13],

$$\mathrm{RV}_t = \tfrac{1}{2}\,(\ln H_t - \ln L_t)^2 - (2\ln 2 - 1)\,(\ln C_t - \ln O_t)^2, \tag{1}$$

and models forecast $y_t = \ln \mathrm{RV}_{t+1}$ at the close of day t. No intraday data are used.

Data-quality repair. The audit found that some Yahoo FX bars are invalid: 36 bars with high $\leq$ low (no real quote) and 469 bars whose open or close lies outside the day's range. Uncorrected, the range estimator reads such a bar as zero or negative variance (floored at 10^{-12}, $\ln \mathrm{RV} \approx -27.6$), and a single such row adds a QLIKE of about 15, enough to distort pooled results. Stale bars are now dropped and out-of-range open/close values are clipped into $[L_t, H_t]$ (table 3).

Table 3: Data universe and data-quality repairs (Yahoo daily OHLC as used by the audited benchmark).

Asset	Class	Daily bars	First	Last	Stale bars dropped	Bars clipped
SPY	Equity / ETF	4,925	2007-01-03	2026-07-31	0	0
QQQ	Equity / ETF	4,925	2007-01-03	2026-07-31	0	0
IWM	Equity / ETF	4,925	2007-01-03	2026-07-31	0	0
AAPL	Equity / ETF	4,925	2007-01-03	2026-07-31	0	0
MSFT	Equity / ETF	4,925	2007-01-03	2026-07-31	0	0
JPM	Equity / ETF	4,925	2007-01-03	2026-07-31	0	0
XOM	Equity / ETF	4,925	2007-01-03	2026-07-31	0	0
JNJ	Equity / ETF	4,925	2007-01-03	2026-07-31	0	0
EURUSD	FX	5,078	2007-01-01	2026-07-31	14	121
USDJPY	FX	5,078	2007-01-01	2026-07-31	5	255
GBPUSD	FX	5,090	2007-01-01	2026-07-31	17	93
TLT	Breadth (not trained on)	4,925	2007-01-03	2026-07-31	0	0
GLD	Breadth (not trained on)	4,925	2007-01-03	2026-07-31	0	0

Stale bar: high $\leq$ low (open = high = low = close, no real quote). Clipped bar: open or close outside [low, high]; clipped into the range. Only the three FX pairs were affected.

Features. The neural model receives 15 features per day, all known at the close of day t: HAR components (daily, weekly and monthly log RV), log RV, return, absolute return, 5-day return, log squared return, log VIX and its daily change, the 10y−2y Treasury term spread, and signed squared-return (semivariance-style)

terms: log downside squared return at daily, weekly and monthly scale and log upside squared return at daily scale. These are built from daily close-to-close returns, not intraday data.

4 The Meridian model

4.1 Architecture

A context window of the last $L = 32$ days of features $x_{t-31:t}$ is linearly projected to $d = 64$ dimensions and passed through two residual blocks, each a layer-normalized diagonal linear state-space layer (in the spirit of S4/Mamba [17, 18]):

$$h_\tau = a \odot h_{\tau-1} + b \odot W_{\text{in}} u_\tau, \qquad o_\tau = W_{\text{out}} h_\tau + W_{\text{skip}} u_\tau, \qquad a = \sigma(\alpha) \in (0,1)^{64}, \tag{2}$$

with a 64-dimensional state. The final belief state h_t feeds:

- a volatility head (two-layer MLP) giving the log-variance forecast f_t;
- auxiliary heads for 5- and 22-day mean log RV;
- a JEPA predictor (MLP) that predicts the embedding of the next window produced by an exponential-moving-average target encoder [2]; its prediction error is a "surprise" energy;
- a HAR skip path: a linear map from $(\text{har}_d, \text{har}_w, \text{har}_m)$ added to every volatility head.

4.2 Training objective

The volatility head is trained on the exact QLIKE loss, which is robust to noise in the variance proxy [28]:

$$\mathcal{L}_{\text{QLIKE}}(f_t, y_t) = e^{\,y_t - f_t} - (y_t - f_t) - 1. \tag{3}$$

The total loss is $\mathcal{L} = \mathcal{L}_{\text{QLIKE}} + 1.0 \cdot \mathcal{L}_{\text{JEPA}} + 0.5 \cdot \mathcal{L}_{\text{SIGReg}} + 0.3 \sum_{h \in \{5,22\}} \mathcal{L}^{(h)}_{\text{QLIKE}}$, where $\mathcal{L}_{\text{SIGReg}}$ is a sketched isotropic-Gaussian regularizer on h_t that prevents representation collapse (matching the first four moments of random 1-D projections to those of $\mathcal{N}(0,1)$, in the spirit of LeJEPA [3]).

4.3 HAR-anchored initialization (configuration "M0")

Following HARNet [31], the HAR skip path is initialized in closed form from the fold's own pooled HAR regression, with the window scaler folded in, so that *before any gradient step* the network's forecast equals the HAR-RV forecast; training then only learns corrections. Each fold trains an ensemble of 5 seeds (AdamW, learning rate 10^{-3}, weight decay 10^{-4}, batch 256, 8 epochs, gradient clipping at 1.0) and averages their log-variance outputs.

5 Benchmark models

Every model forecasts $\ln \text{RV}_{t+1}$ at the close of t and is refit in every fold on training data only. Log models include a Jensen correction $\frac{1}{2}\text{Var}(\text{residual})$ estimated on training data; level models (EWMA, GARCH family) are scaled by a training-only ratio of mean RV to mean forecast.

Not tested. HARQ [6], jump-robust HAR-CJ [1] and Realized GARCH [21] require intraday data and were omitted rather than approximated. Published results from other papers are not tabulated next to ours because they use different assets, periods and variance proxies; every comparison in this report is on identical data and rows.

Algorithm 1 Meridian training and forecasting for one walk-forward fold

Require: training rows with label end $<$ test start; test windows

Fit feature mean/std on training rows only
Fit pooled HAR-OLS $\hat{\beta}$ on training rows (core assets)
for seed $s = 1, \ldots, 5$ **do**
    Initialize network; set HAR skip path from $\hat{\beta}$ so that $f_t^{(s)} \equiv \mathrm{HAR}_t$
    for epoch $= 1, \ldots, 8$ **do**
        for mini-batch B **do**
            $\mathcal{L} \leftarrow$ QLIKE + JEPA + SIGReg + auxiliary QLIKE ▷ §4
            AdamW step; update EMA target encoder
        end for
    end for
end for
return $f_t = \frac{1}{5}\sum_s f_t^{(s)}$ for every test window

Table 4: Volatility models compared (15 plus Meridian and combinations).

Family	Model	Specification
Naive	Random walk	$\ln \mathrm{RV}_t$ plus offset
	22-day mean	log of 22-day mean RV plus offset
	EWMA (RiskMetrics)	$\lambda = 0.94$ on RV [32]
GARCH	GARCH(1,1)	on returns [5]
	GJR-GARCH(1,1)	asymmetric [15]
	EGARCH(1,1)	exponential [26]
HAR	HAR-RV per asset	reference [8]
	HAR-RV pooled	one regression across assets
	LHAR	+ leverage (negative-return) terms [9]
	SHAR (approx.)	daily-sign split of RV; approximation of [29] (no intraday data)
	HAR-X	+ log VIX
ML	Ridge	HAR, leverage and VIX features + asset fixed effects
	Random forest	HAR, semivariance, return and VIX features
	LightGBM	gradient boosting [23], early stopping on a purged slice
Deep	TimeMixer	multiscale mixing network [33], compact reimplementation
Combination	Granger–Ramanathan	walk-forward linear combination of HAR, LightGBM and Meridian [16]

6 Evaluation protocol

Walk-forward. Expanding window; the first test block starts after 1260 trading days; test blocks of 378 days (12 folds); an embargo of 22 days between training and test; training rows are purged so that no label window ends on or after the test start. All scalers are fit on training rows only.

Calibration and combination. Every model receives the same walk-forward affine recalibration $\hat{\sigma}_t^2 = \exp(a + b f_t)$, fit per asset on past forecasts only, with an embargo of 22/30/45 calendar days for the 1/5/22-day targets. The combination is a walk-forward Granger–Ramanathan regression fit the same way.

Development / holdout split. Development = test dates before 2019-01-01 (2012-06-14 to 2018-12-31, 1,901 dates); holdout = 2019-01-01 to 2026-06-29 (2,168 dates). All model choices use development data only; the holdout is reported once.

Statistics. Pooled QLIKE (primary); RMSE of log RV; Mincer–Zarnowitz R^2 and time-series Spearman IC, both averaged over assets; directional hit rate. Significance uses the *date-level* loss differential $d_t = \bar{\ell}_t^m - \bar{\ell}_t^{\text{ref}}$ (mean over assets on date t), because same-date losses across assets are strongly correlated:

$$\text{DM} = \frac{\bar{d}}{\sqrt{\hat{\Omega}/T}}, \qquad \hat{\Omega} = \hat{\gamma}_0 + 2\sum_{k=1}^{q}\Big(1 - \tfrac{k}{q+1}\Big)\hat{\gamma}_k, \quad q = \lfloor 4(T/100)^{2/9} \rfloor = 7, \tag{4}$$

one-sided, Student-t reference [10, 27]. 95% intervals for the percentage change come from a moving-block bootstrap over dates (block 22); the Model Confidence Set [20] uses date-level losses at 90%.

Algorithm 2 Audited evaluation pipeline

for fold $k = 1, \ldots, 12$ **do**
  test block ← next 378 dates; cutoff ← test start − 22 dates
  training rows ← dated before cutoff **and** label end before test start
  fit every model on training rows; forecast the test block
  append raw forecasts to `benchmark_vol_predictions.parquet`
end for
Scoring (separate script, reads only the saved forecasts):
  calibrate every model walk-forward; build combinations walk-forward
  restrict each table to one common row support
  compute QLIKE, date-level DM, bootstrap CI, MCS on dev and holdout separately

7 The audit: what went wrong and what was withdrawn

On 2026-10-31 a review of the benchmark code found eight defects (table 5). The most consequential was that significance tests treated roughly 40,000 stacked asset-date rows as independent observations with no autocorrelation correction; because the 11 assets move together on any given day, this overstated significance by many orders of magnitude.

Lesson

Earlier claims that cleared the bar did so partly because of the evaluation defects (inflated significance, a weakened HAR baseline, corrupted FX rows). After the fixes, Meridian still beats HAR, but the evidence is now measured correctly and every number can be recomputed from saved forecasts in seconds (section A). 11 automated tests guard the fixes (table 29).

Table 5: Defects found by the audit and how each was fixed.

Defect	Effect	Fix
DM test on stacked panel rows, HAC lag 0; MCS on stacked rows	p-values grossly overstated	DM and MCS on date-level losses with Newey–West lag
Invalid FX bars (36 stale, 469 out of range)	variance read as ≈ 0; QLIKE dominated by a few rows	drop / clip (section 3)
HAR fit as one pooled regression	reference model handicapped	per-asset HAR as reference
5- and 22-day labels crossing into test blocks	label leakage	purge by label-end date
TimeMixer scaled with full-sample statistics	look-ahead	training-only scaling
EWMA/GARCH forecasts one day stale	baselines unfairly weak	true one-step-ahead forecasts
Raw and calibrated tables on different rows	sample change mixed with model change	one common row support per table
No saved predictions; numbers copied by hand	nothing re-checkable	predictions saved; tables generated

Table 6: History of headline claims made before the audit. All are **withdrawn**: they were produced by the defective harness and are not reproducible from this repository. Before the audit, "$+x\%$" meant a QLIKE *reduction*.

Source	Claim as reported at the time	Status
Amendment 1 (07-29)	Meridian-MSE $+2.51\%$ vs HAR, $p = 0.0036$; Meridian-QLIKE $+3.15\%$, $p = 0.0020$ (below the 5% bar)	withdrawn
RESULTS.md "FINAL"	5-seed ensemble $+6.27\%$ vs HAR, DM $p = 1.17 \times 10^{-9}$, "primary claim met"	withdrawn
RESULTS.md	CF-JEPA EMA readout, ensembled: $+6.78\%$ vs HAR	withdrawn
RESULTS.md (Am. 4)	real neural core in the benchmark "still doesn't beat HAR/EWMA"	superseded
Commit `ab041b0` (Am. 5)	"-8.7% QLIKE, $p < 0.001$"; elsewhere $p \approx 10^{-14}$	withdrawn
RESULTS.md (regimes)	regime persistence vs HMM: not met; "economic value" win	not re-validated
RESULTS.md / RESEARCH.md	trading-strategy experiments did not reach the Sharpe target (≈ 1.5) sought	not reproduced

8 Results: volatility forecasting

8.1 Headline comparison

Table 7 and fig. 1 give the holdout comparison of all calibrated models. Calibrated Meridian (QLIKE 0.3045) improves on per-asset HAR (QLIKE 0.3292) by −7.5% [-11.3, -4.7], p <0.001, and wins on 11/11 assets. The walk-forward combination of HAR, LightGBM and Meridian reaches −8.2% [-12.4, -5.2], a small gain over Meridian alone. Among non-Meridian models the best are LightGBM (−5.3%), HAR-X (−5.0%) and Ridge (−4.9%). TimeMixer (+3.0%), the GARCH family (+12.2% to +14.9%), EWMA (+14.1%) and the random walk (+18.0%) are worse than HAR.

Table 7: Volatility forecasting, holdout 2019-01-01 to 2026-06-29 (2,168 dates × 11 assets, 20,906 forecasts per model). All models recalibrated identically; reference = per-asset HAR-RV. Sorted by QLIKE (lower is better).

Model	Family	QLIKE	vs HAR [95% CI]	DM p	MZ R^2	IC	Hit %	Won	MCS
GR combo: HAR + LightGBM + Meridian M0 (pooled HAR anchor)	Combo	0.3022	−8.2% [-12.4, -5.2]	<0.001	0.413	0.594	68.3	11/11	✓
Meridian M0 (pooled HAR anchor)	Meridian	0.3045	−7.5% [-11.3, -4.7]	<0.001	0.410	0.591	68.1	11/11	
GR combo: HAR + LightGBM	Combo	0.3105	−5.7% [-9.3, -3.1]	<0.001	0.400	0.584	67.8	10/11	
LightGBM	ML	0.3118	−5.3% [-9.3, -2.5]	<0.001	0.399	0.583	67.9	10/11	
HAR-X (+VIX)	HAR	0.3126	−5.0% [-7.7, -2.9]	<0.001	0.391	0.578	66.9	10/11	
Ridge	ML	0.3132	−4.9% [-8.7, -2.1]	<0.001	0.397	0.578	67.2	9/11	
Random forest	ML	0.3171	−3.7% [-6.8, -1.4]	<0.001	0.388	0.573	67.6	11/11	
LHAR (leverage)	HAR	0.3204	−2.7% [-5.4, -0.5]	0.011	0.386	0.569	68.0	10/11	
SHAR (daily-sign semivariance)	HAR	0.3261	−0.9% [-1.6, -0.4]	0.004	0.372	0.566	67.2	8/11	
HAR-RV, pooled	HAR	0.3291	−0.0% [-0.4, +0.3]	0.285	0.368	0.563	66.8	4/11	
HAR-RV, per asset (reference)	HAR	0.3292	+0.0%	—	0.367	0.562	66.9	0/11	
TimeMixer	Deep	0.3392	+3.0% [+0.3, +6.1]	0.957	0.345	0.542	67.0	3/11	
EGARCH(1,1)	GARCH	0.3695	+12.2% [+8.1, +16.8]	1.000	0.321	0.515	68.2	0/11	
EWMA RiskMetrics λ=0.94	Naive	0.3758	+14.1% [+7.9, +21.8]	1.000	0.298	0.513	66.8	0/11	
GJR-GARCH(1,1)	GARCH	0.3758	+14.1% [+9.9, +18.9]	1.000	0.315	0.504	68.1	0/11	
GARCH(1,1)	GARCH	0.3782	+14.9% [+11.0, +18.6]	1.000	0.293	0.490	67.7	0/11	
22-day mean RV	Naive	0.3833	+16.4% [+9.1, +25.9]	1.000	0.283	0.501	67.0	0/11	
Random walk (log RV_t)	Naive	0.3884	+18.0% [+12.9, +24.7]	1.000	0.266	0.472	62.3	0/11	

Green: Meridian rows. Red: worse than HAR. DM p: one-sided Diebold–Mariano on the date-level loss differential with Newey–West HAC. CI: moving-block bootstrap over dates. Won: assets on which the model's QLIKE is below HAR's. MCS: in the 90% Model Confidence Set.

Hit rate is not evidence of skill here. Calibrated models' directional hit rates range only from 62.3% to 68.3%: volatility is persistent, so predicting whether tomorrow's variance is above today's is easy for any reasonable model. Hit rate is reported but not used to rank models.

8.2 Conservative check against the strongest HAR

On this holdout raw per-asset HAR (QLIKE 0.3249) scores better than its calibrated version (+1.3% worse), so the most demanding comparison is against raw HAR (table 8). Raw Meridian improves on it by −6.8% [-9.7, -4.5], p <0.001; the combination by −7.0% [-9.5, -4.9].

8.3 Stability over time, assets and market conditions

Meridian beats raw HAR in 14 of 15 calendar years (2012–2026; the exception is 2026, a partial year, at +0.04%; fig. 2 and table 10). On the holdout it beats HAR on every asset, from −12.6% (QQQ) to −1.1% (USDJPY) (fig. 4 and table 11). The gain is largest in high-VIX periods (−14.1% in the stress tercile vs −4.1% in the calm tercile; table 12). On individual days the edge is noisy: Meridian has the lower loss on 61.2% of holdout days, on 86.6% of 20-day windows, 96.5% of 63-day windows and 100.0% of 252-day windows (fig. 3).

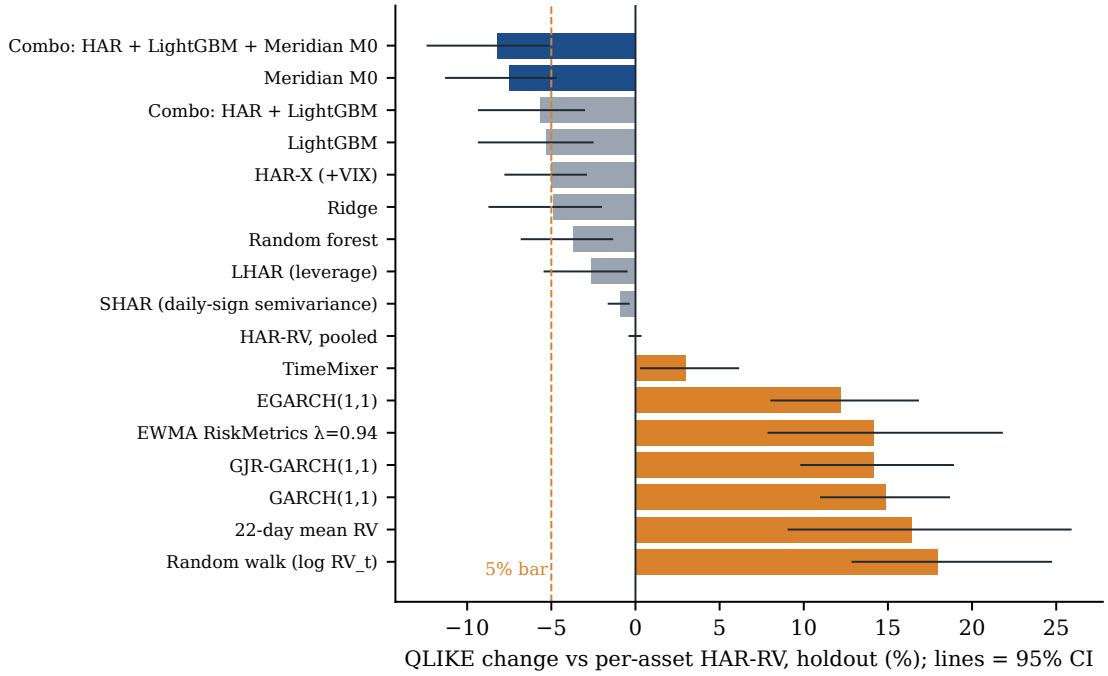


Figure 1: Holdout QLIKE change vs per-asset HAR-RV with 95% block-bootstrap intervals. Blue: Meridian; grey: better than HAR; orange: worse than HAR. Dashed line: the pre-registered 5% bar.

Table 8: Conservative check: holdout against raw per-asset HAR, the strongest HAR form.

Model	QLIKE	vs raw HAR [95% CI]	DM p	Won
GR combo: HAR + LightGBM + Meridian M0 (pooled HAR anchor)	0.3022	−7.0% [-9.5, -4.9]	<0.001	11/11
Meridian M0 (pooled HAR anchor) — raw	0.3029	−6.8% [-9.7, -4.5]	<0.001	11/11
Meridian M0 (pooled HAR anchor) — calibrated	0.3045	−6.3% [-8.6, -4.3]	<0.001	11/11
GR combo: HAR + LightGBM	0.3105	−4.4% [-6.6, -2.7]	<0.001	10/11
LightGBM — calibrated	0.3118	−4.0% [-6.5, -2.3]	<0.001	10/11
HAR-RV, per asset — raw	0.3249	+0.0%	—	0/11
HAR-RV, per asset — calibrated	0.3292	+1.3% [-0.2, +3.9]	0.914	0/11

Table 9: Raw (uncalibrated) forecasts on the holdout, reference = raw per-asset HAR-RV.

Model (raw, uncalibrated)	Family	QLIKE	vs raw HAR [95% CI]	DM p	Won
Meridian M0 (pooled HAR anchor)	Meridian	0.3029	−6.8% [-9.7, -4.5]	<0.001	11/11
HAR-X (+VIX)	HAR	0.3119	−4.0% [-5.7, -2.5]	<0.001	10/11
Ridge	ML	0.3133	−3.6% [-6.2, -1.3]	<0.001	10/11
LightGBM	ML	0.3137	−3.4% [-6.3, -1.4]	0.003	10/11
LHAR (leverage)	HAR	0.3159	−2.8% [-4.8, -1.3]	0.001	10/11
Random forest	ML	0.3195	−1.7% [-4.0, +0.2]	0.039	8/11
SHAR (daily-sign semivariance)	HAR	0.3218	−0.9% [-1.6, -0.4]	0.008	7/11
HAR-RV, per asset (reference)	HAR	0.3249	+0.0%	—	0/11
HAR-RV, pooled	HAR	0.3285	+1.1% [+0.4, +1.6]	1.000	2/11
TimeMixer	Deep	0.3570	+9.9% [+6.5, +13.9]	1.000	0/11
EWMA RiskMetrics λ=0.94	Naive	0.3627	+11.6% [+6.2, +18.0]	1.000	2/11
22-day mean RV	Naive	0.3779	+16.3% [+10.6, +22.4]	1.000	0/11
EGARCH(1,1)	GARCH	0.4218	+29.8% [+25.1, +34.4]	1.000	0/11
GJR-GARCH(1,1)	GARCH	0.4230	+30.2% [+25.8, +34.2]	1.000	0/11
GARCH(1,1)	GARCH	0.4307	+32.6% [+28.2, +37.0]	1.000	0/11
Random walk (log RV_t)	Naive	0.5205	+60.2% [+46.9, +69.5]	1.000	0/11

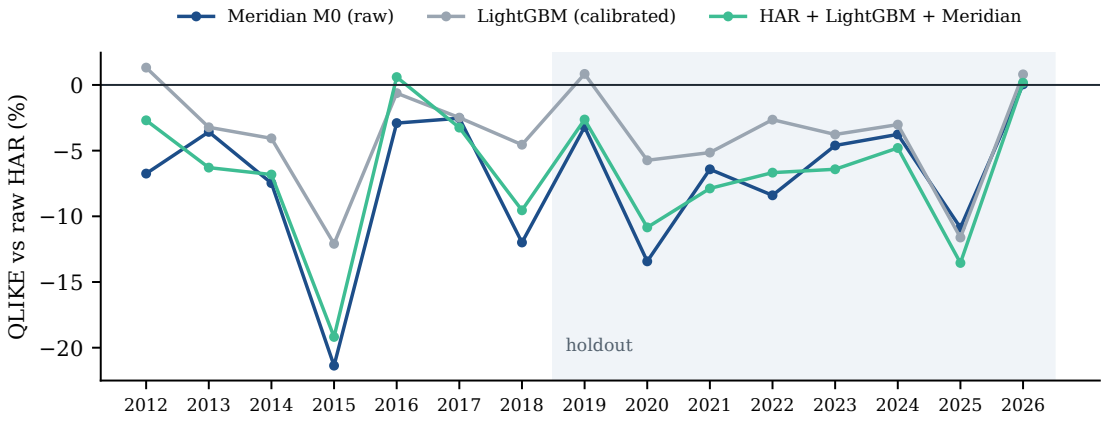


Figure 2: QLIKE change vs raw per-asset HAR by calendar year (shaded: holdout).

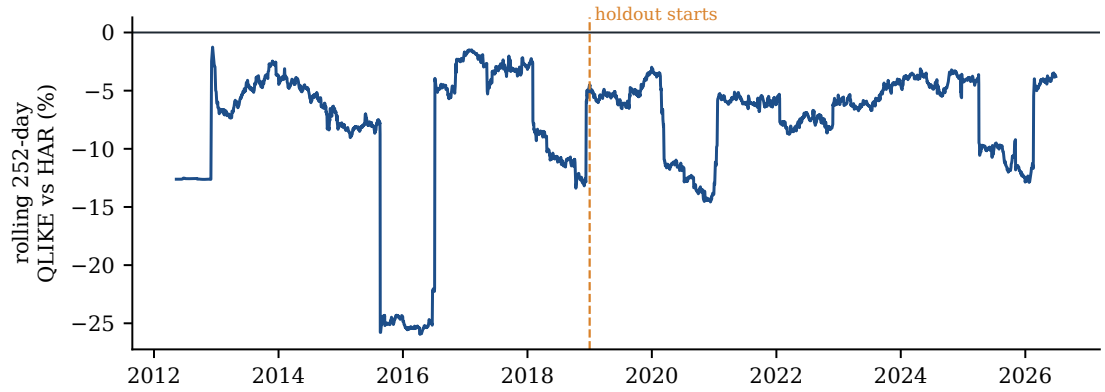


Figure 3: Rolling 252-day QLIKE difference of raw Meridian vs raw per-asset HAR, as a percentage of HAR's loss, over the full out-of-sample period.

Table 10: QLIKE change vs raw per-asset HAR by calendar year (all out-of-sample years).

Year	Meridian (raw)	LightGBM (cal.)	HAR+LGBM+Meridian	Period
2012	−6.7%	+1.3%	−2.7%	development
2013	−3.6%	−3.2%	−6.3%	development
2014	−7.5%	−4.1%	−6.8%	development
2015	−21.4%	−12.1%	−19.2%	development
2016	−2.9%	−0.6%	+0.6%	development
2017	−2.5%	−2.5%	−3.2%	development
2018	−12.0%	−4.5%	−9.5%	development
2019	−3.2%	+0.8%	−2.6%	holdout
2020	−13.4%	−5.7%	−10.8%	holdout
2021	−6.4%	−5.1%	−7.9%	holdout
2022	−8.4%	−2.6%	−6.7%	holdout
2023	−4.6%	−3.8%	−6.4%	holdout
2024	−3.8%	−3.0%	−4.8%	holdout
2025	−10.9%	−11.6%	−13.5%	holdout
2026	+0.0%	+0.8%	+0.2%	holdout

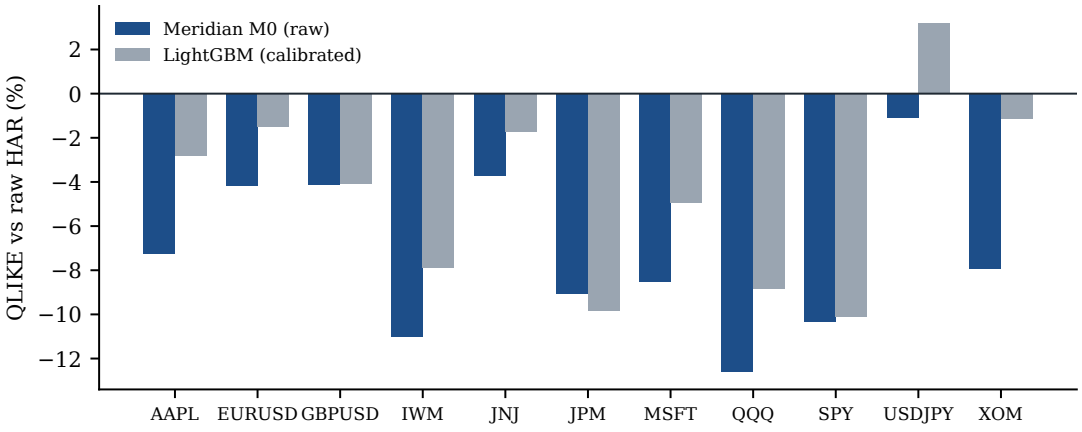


Figure 4: Holdout QLIKE change vs raw per-asset HAR, by asset.

Table 11: Holdout QLIKE change vs raw per-asset HAR, by asset.

Asset	Meridian (raw)	Meridian (cal.)	LightGBM (cal.)	HAR+LGBM+Meridian
AAPL	−7.3%	−7.1%	−2.8%	−6.9%
EURUSD	−4.2%	−3.3%	−1.5%	−3.0%
GBPUSD	−4.1%	−1.3%	−4.1%	−4.1%
IWM	−11.0%	−10.6%	−7.9%	−10.1%
JNJ	−3.7%	−3.9%	−1.7%	−4.2%
JPM	−9.1%	−9.0%	−9.9%	−11.4%
MSFT	−8.5%	−8.4%	−5.0%	−8.3%
QQQ	−12.6%	−12.7%	−8.8%	−13.7%
SPY	−10.3%	−10.4%	−10.1%	−12.5%
USDJPY	−1.1%	−1.0%	+3.2%	−1.0%
XOM	−7.9%	−6.6%	−1.1%	−6.8%

Table 12: QLIKE change vs calibrated per-asset HAR by VIX tercile, full out-of-sample period (2012-08-13 to 2026-06-28); DM p in brackets.

Model	Calm	Normal	Stress
Meridian M0 (pooled HAR anchor)	−4.1% (<0.001)	−5.2% (<0.001)	−14.1% (0.009)
GR combo: HAR + LightGBM + Meridian M0 (pooled HAR anchor)	−4.9% (<0.001)	−6.4% (0.034)	−15.3% (0.005)
LightGBM	−2.9% (<0.001)	−4.1% (0.203)	−10.9% (0.004)
HAR-X (+VIX)	−1.8% (<0.001)	−2.8% (<0.001)	−9.6% (0.004)
Ridge	−1.6% (<0.001)	−2.6% (<0.001)	−10.7% (0.010)
LHAR (leverage)	−1.2% (<0.001)	−2.2% (<0.001)	−7.1% (0.035)
TimeMixer	+2.6% (0.975)	+1.5% (0.869)	+3.3% (0.931)
GJR-GARCH(1,1)	+9.6% (1.000)	+8.1% (1.000)	+12.9% (0.984)
EWMA RiskMetrics λ=0.94	+6.2% (1.000)	+9.0% (1.000)	+26.9% (0.999)

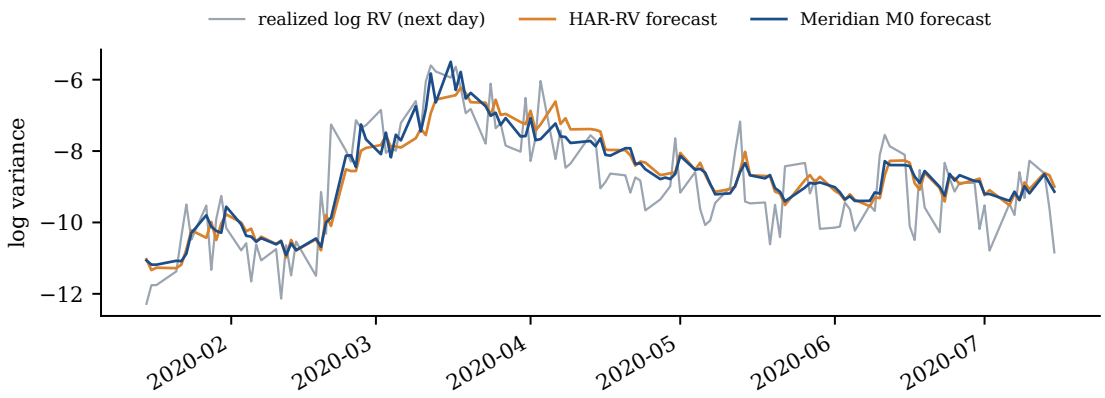


Figure 5: Illustration on real data: SPY, January–July 2020. Next-day realized log variance with the HAR-RV and Meridian forecasts made the previous evening.

8.4 Longer horizons and out-of-universe assets

At 5 days Meridian's auxiliary head improves on a horizon-specific HAR by −9.6% (p <0.001); at 22 days the difference, −0.9%, is not significant (p 0.356) (table 13). Only HAR and Meridian were compared at these horizons. On TLT and GLD, which the pooled models never saw in training, calibrated Meridian improves on HAR by −7.1% [-12.4, -2.1], p 0.002 (table 14).

Table 13: Longer horizons (holdout, calibrated): Meridian auxiliary heads vs per-asset HAR fit to the same horizon.

Horizon	HAR QLIKE	Meridian QLIKE	vs HAR [95% CI]	DM p	Won
5 trading days	0.1900	0.1718	−9.6% [-14.5, -4.9]	<0.001	11/11
22 trading days	0.2086	0.2067	−0.9% [-4.5, +6.7]	0.356	7/11

Table 14: Out-of-universe check: TLT (long Treasuries) and GLD (gold), holdout. Pooled models (Meridian, Ridge, random forest, LightGBM, TimeMixer, pooled HAR) never trained on these assets.

Model (calibrated)	QLIKE	vs HAR [95% CI]	DM p
Meridian M0 (pooled HAR anchor)	0.3086	−7.1% [-12.4, -2.1]	0.002
HAR-X (+VIX)	0.3265	−1.7% [-6.6, +1.9]	0.181
HAR-RV, pooled	0.3267	−1.6% [-4.5, +1.1]	0.110
Ridge	0.3300	−0.6% [-7.0, +4.8]	0.405
HAR-RV, per asset (reference)	0.3320	+0.0%	–
SHAR (daily-sign semivariance)	0.3323	+0.1% [-0.4, +0.8]	0.613
LHAR (leverage)	0.3346	+0.8% [-0.1, +1.8]	0.931
Random forest	0.3377	+1.7% [-2.1, +5.9]	0.822
LightGBM	0.3473	+4.6% [-0.4, +10.4]	0.961
EWMA RiskMetrics λ=0.94	0.3594	+8.2% [+4.2, +12.8]	1.000
22-day mean RV	0.3676	+10.7% [+4.5, +17.7]	1.000
TimeMixer	0.3716	+11.9% [+6.5, +17.5]	1.000
GJR-GARCH(1,1)	0.3861	+16.3% [+6.4, +29.5]	0.999
EGARCH(1,1)	0.3877	+16.8% [+5.6, +32.2]	0.995
GARCH(1,1)	0.3899	+17.4% [+7.0, +31.2]	0.999
Random walk (log RV_t)	0.4001	+20.5% [+11.4, +29.9]	1.000

8.5 Development period and the improvement attempt

Three architectural changes were proposed to improve Meridian and evaluated only on the development period (table 15 and fig. 6): M1 replaces the pooled HAR anchor with a per-asset HAR offset plus a zero-initialized correction; M2 adds a learned asset embedding; M3 adds early stopping. None beat the existing configuration M0 (development QLIKE: M0 0.3404, M1 0.3432, M2 0.3430, M3 0.3477). This is

recorded as a negative result; M0 was kept and the holdout was evaluated once. Table 16 gives the full development-period table of the final run (Meridian −8.8%).

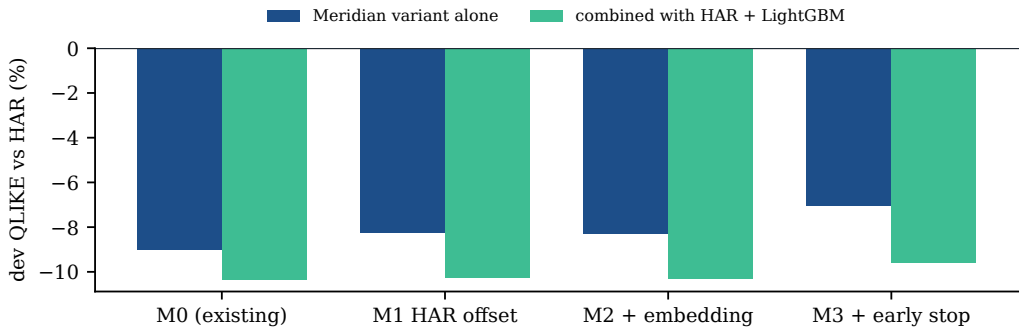


Figure 6: Development-period ablation: QLIKE change vs calibrated HAR for the existing configuration (M0) and the three proposed changes, alone and in combination.

Table 15: Development-period ablation of the proposed Meridian improvements (3 seeds each). The selection rule picks the lowest development QLIKE.

Candidate	Change tested	Dev QLIKE	vs HAR
GR combo: HAR + LightGBM + Meridian M0 (pooled HAR anchor)	pooled HAR-anchored initialization (existing)	0.3353	−10.4%
GR combo: HAR + LightGBM + Meridian M2 (M1 + asset embedding)	M1 + learned asset embedding	0.3354	−10.3%
GR combo: HAR + LightGBM + Meridian M1 (per-asset HAR offset)	per-asset HAR offset + zero-init correction	0.3356	−10.3%
GR combo: HAR + LightGBM + Meridian M3 (M1 + early stopping)	M1 + early stopping (purged validation slice)	0.3381	−9.6%
Meridian M0 (pooled HAR anchor)	pooled HAR-anchored initialization (existing)	0.3404	−9.0%
Meridian M2 (M1 + asset embedding)	M1 + learned asset embedding	0.3430	−8.3%
Meridian M1 (per-asset HAR offset)	per-asset HAR offset + zero-init correction	0.3432	−8.3%
Meridian M3 (M1 + early stopping)	M1 + early stopping (purged validation slice)	0.3477	−7.1%
HAR-RV, per asset (reference)	reference	0.3741	+0.0%

Table 16: Development period 2012-06-14 to 2018-12-31 (final 5-seed run).

Model (calibrated)	QLIKE	vs HAR [95% CI]	DM p	Won
GR combo: HAR + LightGBM + Meridian M0 (pooled HAR anchor)	0.3356	−10.3% [-16.3, -5.4]	0.017	11/11
Meridian M0 (pooled HAR anchor)	0.3410	−8.8% [-15.2, -3.8]	0.016	11/11
GR combo: HAR + LightGBM	0.3461	−7.5% [-11.2, -4.1]	0.034	11/11
LightGBM	0.3469	−7.3% [-10.9, -4.0]	0.023	10/11
Random forest	0.3515	−6.0% [-9.7, -3.1]	0.042	10/11
Ridge	0.3528	−5.7% [-10.4, -2.3]	0.038	10/11
HAR-X (+VIX)	0.3552	−5.0% [-8.6, -2.3]	0.025	11/11
LHAR (leverage)	0.3561	−4.8% [-9.0, -1.6]	0.040	11/11
SHAR (daily-sign semivariance)	0.3696	−1.2% [-2.6, -0.0]	0.061	9/11
HAR-RV, pooled	0.3728	−0.3% [-0.6, -0.1]	0.011	7/11
HAR-RV, per asset (reference)	0.3741	+0.0%	—	0/11
TimeMixer	0.3818	+2.1% [-0.2, +4.3]	0.966	3/11
GJR-GARCH(1,1)	0.3987	+6.6% [-0.4, +12.0]	0.958	4/11
EGARCH(1,1)	0.4004	+7.0% [+1.9, +11.4]	0.996	2/11
Random walk (log RV_t)	0.4028	+7.7% [+0.0, +13.6]	0.993	1/11
GARCH(1,1)	0.4218	+12.8% [+8.9, +16.1]	1.000	0/11
EWMA RiskMetrics λ=0.94	0.4319	+15.5% [+8.5, +23.7]	0.995	1/11
22-day mean RV	0.4349	+16.3% [+9.3, +24.3]	0.996	0/11

9 Results: tail risk (Value-at-Risk and Expected Shortfall)

9.1 Question and models

Does Meridian's volatility forecast produce better next-day VaR and ES of close-to-close returns than the standard risk models? Levels $\alpha \in \{1\%, 2.5\%, 5\%\}$; zero conditional mean for all models; same folds as section 8. The design (Amendment 7) was declared before any result was computed.

Table 17: Tail-risk models compared.

Model	Specification
Historical simulation	empirical quantile and tail mean of the last 500 returns
RiskMetrics	EWMA variance ($\lambda = 0.94$), normal
GARCH(1,1)-normal / -t	parametric VaR/ES
GJR-GARCH-skewed-t	Hansen skewed-t [19]
GJR-GARCH + FHS (reference)	filtered historical simulation [4]
GJR-GARCH + EVT	GPD on standardized residuals [25]
CAViaR (asymmetric slope)	quantile autoregression [11]; VaR only
HAR / LightGBM / Meridian + FHS	returns filtered by the saved out-of-sample volatility forecast
HAR / Meridian + EVT	as above with a GPD tail
Meridian + FHS + ACI	adaptive conformal level [14], $\gamma = 0.005$

Algorithm 3 Volatility-filtered VaR/ES (FHS and EVT variants), forecast at origin t

Require: volatility forecasts $\hat{\sigma}_{s+1|s}$ and returns r_{s+1} for origins $s < t$

$z_s \leftarrow r_{s+1}/\hat{\sigma}_{s+1|s}$ for the last ≤ 1000 origins $s < t$ (need ≥ 250)

if FHS **then**

  $q_\alpha \leftarrow$ empirical α-quantile of z; $e_\alpha \leftarrow$ mean of z below q_α

else ▷ EVT, refit every 21 days

  $u \leftarrow$ 0.90-quantile of losses $-z$; fit GPD(ξ, β) to exceedances over u

  $q_\alpha \leftarrow -\left[u + \frac{\beta}{\xi}\left(\left(\frac{n\alpha}{N_u}\right)^{-\xi} - 1\right)\right]$; $e_\alpha \leftarrow -\frac{-q_\alpha + \beta - \xi u}{1-\xi}$

end if

return $\mathrm{VaR}_t = \hat{\sigma}_{t+1|t}\, q_\alpha$, $\mathrm{ES}_t = \hat{\sigma}_{t+1|t}\, e_\alpha$

Scoring. The primary score is the FZ0 loss [12, 30], which is strictly consistent for the pair (VaR, ES):

$$L_{\mathrm{FZ0}}(v, e; r) = -\frac{1}{\alpha e}\,\mathbf{1}\{r \le v\}(v - r) + \frac{v}{e} + \ln(-e) - 1, \qquad e \le v < 0. \tag{5}$$

Differences in FZ0 between models are invariant to each asset's return scale, so assets can be pooled. VaR alone is scored by the quantile loss $(\alpha - \mathbf{1}\{r < v\})(r - v)$ on returns divided by a fixed per-asset scale. Absolute calibration is checked per asset with the Kupiec unconditional-coverage test [24], the Christoffersen conditional-coverage test [7] and the Engle–Manganelli dynamic-quantile (DQ) test; ES with the realized/forecast ratio on violation days. Comparative significance uses the same date-level DM test, bootstrap and MCS as section 6. The Meridian variant (FHS, EVT or FHS+ACI) was chosen on the development period: Meridian + EVT.

9.2 Results

Meridian + EVT has the lowest joint VaR+ES loss on the holdout: FZ0 difference -0.1218 [-0.1533, -0.0949] vs the GJR-GARCH + FHS reference, p <0.001 (table 18 and fig. 7). Its violation rates are 0.96%, 2.49% and 4.95% at the 1%, 2.5% and 5% levels; it passes the Kupiec and Christoffersen tests on 11/11 and 11/11 assets at 1% and 11/11 and 11/11 at 5%, and the DQ test on 10/11 assets at 2.5%; realized tail losses are 0.98

times the forecast ES. At 1% its quantile loss is -9.2% [-12.7, -5.9] vs the reference (p <0.001). Historical simulation is worst ($+26.3\%$ at 1%), and the normal-based models under-cover at 1% (violation rates 1.80% for GARCH-normal and 2.10% for RiskMetrics).

Table 18: Joint VaR + ES at 2.5% (FZ0 loss, lower is better), holdout 2019-01-01 to 2026-06-29 (2,168 dates × 11 assets). Reference: GJR-GARCH + filtered historical simulation.

Model	FZ0	Δ vs reference [95% CI]	DM p	MCS	Viol. %	CC · DQ pass	ES ratio
Meridian + FHS	-3.6394	-0.1226 [-0.1530, -0.0959]	<0.001	✓	2.54	11/11 · 10/11	0.97
Meridian + EVT	-3.6387	-0.1218 [-0.1533, -0.0949]	<0.001	✓	2.49	11/11 · 10/11	0.98
HAR-RV + FHS	-3.6289	-0.1182 [-0.1440, -0.0927]	<0.001	✓	2.58	11/11 · 8/11	0.99
HAR-RV + EVT	-3.6284	-0.1179 [-0.1449, -0.0912]	<0.001	✓	2.57	11/11 · 7/11	0.99
Meridian + FHS + adaptive conformal	-3.6257	-0.1080 [-0.1448, -0.0782]	<0.001		2.47	11/11 · 11/11	0.98
LightGBM vol + FHS	-3.6208	-0.1048 [-0.1306, -0.0796]	<0.001		2.58	11/11 · 10/11	0.98
GJR-GARCH-skewed-t	-3.5294	-0.0013 [-0.0124, +0.0093]	0.404		2.60	10/11 · 10/11	0.99
GJR-GARCH + EVT (McNeil-Frey)	-3.5290	-0.0030 [-0.0054, -0.0007]	0.007		2.67	11/11 · 11/11	0.98
GJR-GARCH + FHS (reference)	-3.5255	—	—		2.71	11/11 · 11/11	0.98
GARCH(1,1)-t	-3.5203	+0.0067 [-0.0097, +0.0234]	0.799		2.87	10/11 · 7/11	1.00
GARCH(1,1)-normal	-3.5049	+0.0179 [-0.0036, +0.0427]	0.951		3.01	9/11 · 7/11	1.14
RiskMetrics EWMA, normal	-3.4538	+0.0626 [+0.0215, +0.1094]	0.999		3.48	4/11 · 1/11	1.16
Historical simulation (500d)	-3.3320	+0.1836 [+0.0660, +0.3735]	0.997		2.80	5/11 · 1/11	1.07

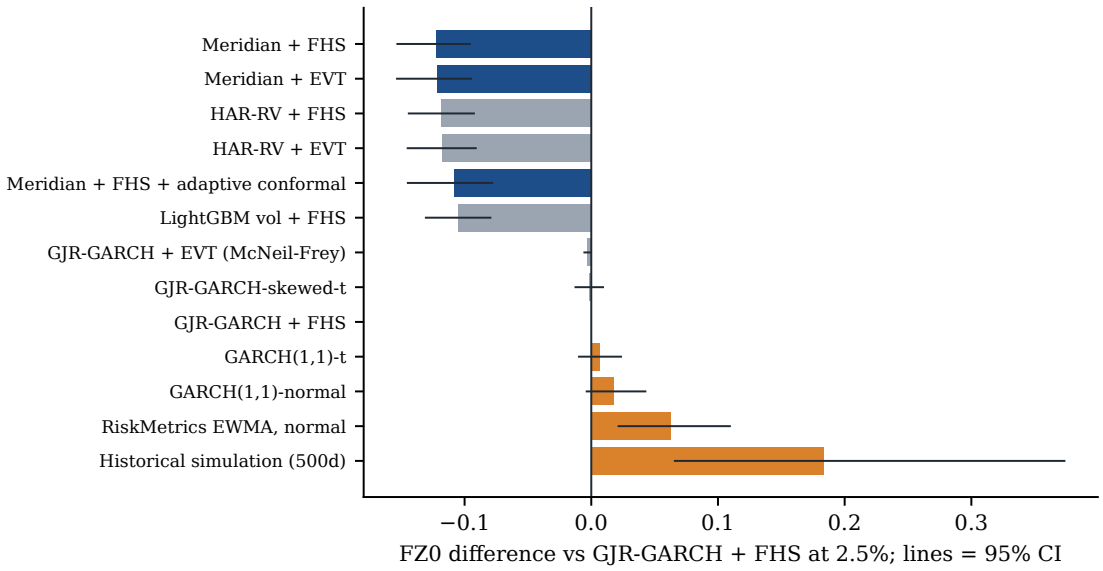


Figure 7: Holdout FZ0 loss difference vs GJR-GARCH + FHS at 2.5%, with 95% intervals.

The honest reading of the tail results

The clear separation is between models that use a range-based realized-volatility forecast (Meridian, HAR, LightGBM) and models that use returns only (GARCH family, RiskMetrics, historical simulation, CAViaR). Against the HAR-filtered tail the Meridian advantage is **not statistically significant** (table 19: p 0.076 at 1%, 0.369 at 2.5%, 0.203 at 5%). On the development period the HAR-filtered tail was in fact slightly better (FZ0 -3.7832 vs -3.7753; table 23). Meridian's tails do cluster less: they pass the DQ test on 10/11 assets vs 8/11 for HAR + FHS.

10 Forward test

A backtest, however carefully protected, is scored on data that existed when the model was designed. The forward test removes that risk.

Table 19: Head-to-head on the holdout: Meridian + EVT vs HAR-RV + FHS (both filtered by a realized-volatility forecast).

Level	FZ0 Δ [95% CI]	DM p	Quantile loss vs HAR-FHS [95% CI]	DM p
1%	-0.0100 [-0.0461, +0.0168]	0.245	−3.1% [-7.4, +0.2]	0.076
2.5%	-0.0037 [-0.0333, +0.0175]	0.369	−2.0% [-6.2, +1.1]	0.159
5%	-0.0013 [-0.0246, +0.0144]	0.439	−1.1% [-4.1, +1.0]	0.203

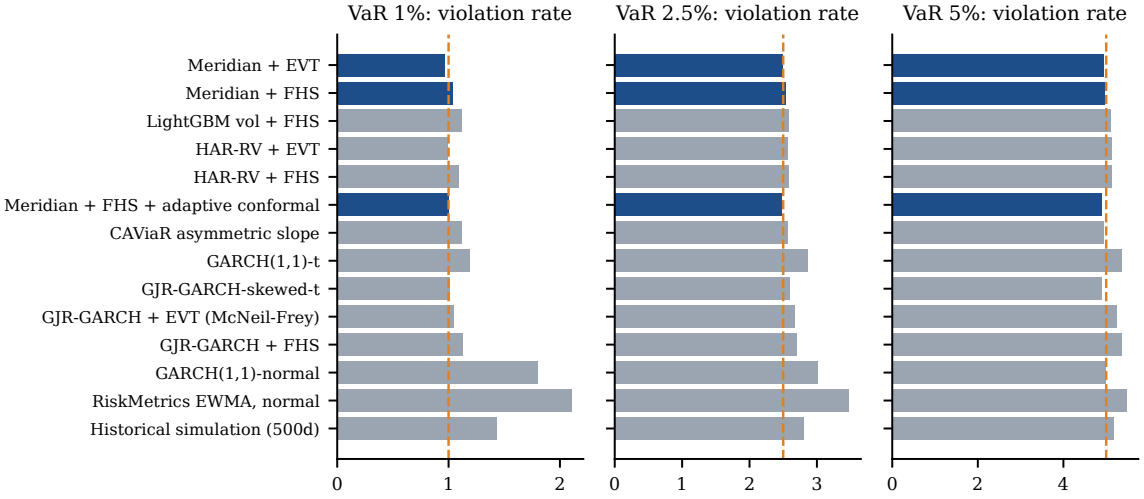


Figure 8: Holdout VaR violation rates by model; dashed lines mark the nominal level.

Table 20: VaR at 1%: quantile loss on returns scaled per asset, holdout.

Model	Loss $\times 10^3$	vs reference [95% CI]	DM p	MCS	Viol. % (target 1)	Kupiec · CC · DQ pass (of 11)
Meridian + EVT	22.27	−9.2% [-12.7, -5.9]	<0.001	✓	0.96	11 · 11 · 10
Meridian + FHS	22.32	−9.0% [-12.5, -5.7]	<0.001	✓	1.04	11 · 11 · 10
LightGBM vol + FHS	22.59	−7.9% [-10.8, -4.7]	<0.001	✓	1.11	11 · 11 · 10
HAR-RV + EVT	22.98	−6.3% [-9.7, -2.5]	<0.001	✓	0.99	11 · 10 · 7
HAR-RV + FHS	22.99	−6.3% [-9.6, -2.7]	<0.001	✓	1.09	11 · 10 · 6
Meridian + FHS + adaptive conformal	23.26	−5.2% [-9.1, -1.5]	<0.001		1.00	11 · 11 · 11
CAViaR asymmetric slope	24.21	−1.3% [-2.9, +0.6]	0.181		1.11	10 · 10 · 10
GARCH(1,1)-t	24.39	−0.6% [-2.7, +1.7]	0.357		1.19	9 · 11 · 8
GJR-GARCH-skewed-t	24.47	−0.3% [-1.4, +1.1]	0.437		1.01	11 · 11 · 11
GJR-GARCH + EVT (McNeil-Frey)	24.51	−0.1% [-0.5, +0.4]	0.411		1.05	11 · 11 · 11
GJR-GARCH + FHS (reference)	24.53	+0.0%	–		1.12	11 · 11 · 10
GARCH(1,1)-normal	25.08	+2.2% [-1.9, +7.3]	0.828		1.80	2 · 2 · 2
RiskMetrics EWMA, normal	26.37	+7.5% [+0.1, +17.1]	0.972		2.10	0 · 0 · 0
Historical simulation (500d)	30.97	+26.3% [+6.7, +57.4]	0.980		1.43	8 · 4 · 0

Table 21: VaR at 2.5%: quantile loss on returns scaled per asset, holdout.

Model	Loss $\times 10^3$	vs reference [95% CI]	DM p	MCS	Viol. % (target 2.5)	Kupiec · CC · DQ pass (of 11)
Meridian + FHS	45.54	−7.4% [-9.6, -5.2]	<0.001	✓	2.54	11 · 11 · 10
Meridian + EVT	45.54	−7.4% [-9.7, -5.1]	<0.001	✓	2.49	11 · 11 · 10
LightGBM vol + FHS	46.10	−6.3% [-8.4, -4.0]	<0.001		2.58	11 · 11 · 10
Meridian + FHS + adaptive conformal	46.11	−6.2% [-9.4, -3.5]	<0.001		2.47	11 · 11 · 11
HAR-RV + FHS	46.46	−5.5% [-8.0, -2.7]	<0.001	✓	2.58	11 · 11 · 8
HAR-RV + EVT	46.56	−5.3% [-8.1, -2.2]	<0.001	✓	2.57	11 · 11 · 7
GJR-GARCH-skewed-t	48.89	−0.6% [-2.3, +0.7]	0.231		2.60	10 · 10 · 10
GJR-GARCH + EVT (McNeil-Frey)	49.04	−0.3% [-0.5, -0.1]	0.005		2.67	11 · 11 · 11
CAViaR asymmetric slope	49.09	−0.2% [-1.6, +1.3]	0.730		2.57	8 · 9 · 9
GJR-GARCH + FHS (reference)	49.18	+0.0%	—		2.71	11 · 11 · 11
GARCH(1,1)-normal	49.36	+0.4% [-1.0, +2.1]	0.684		3.01	6 · 9 · 7
GARCH(1,1)-t	49.41	+0.5% [-0.8, +1.9]	0.780		2.87	8 · 10 · 7
RiskMetrics EWMA, normal	50.84	+3.4% [+0.4, +7.2]	0.979		3.48	3 · 4 · 1
Historical simulation (500d)	56.55	+15.0% [+3.9, +31.6]	0.987		2.80	11 · 5 · 1

Table 22: VaR at 5%: quantile loss on returns scaled per asset, holdout.

Model	Loss $\times 10^3$	vs reference [95% CI]	DM p	MCS	Viol. % (target 5)	Kupiec · CC · DQ pass (of 11)
Meridian + EVT	77.07	−5.1% [-6.7, -3.6]	<0.001	✓	4.95	11 · 11 · 9
Meridian + FHS	77.17	−5.0% [-6.5, -3.5]	<0.001		4.97	11 · 11 · 9
Meridian + FHS + adaptive conformal	77.74	−4.3% [-6.2, -2.5]	<0.001		4.90	11 · 11 · 9
LightGBM vol + FHS	77.78	−4.2% [-5.6, -2.9]	<0.001		5.11	11 · 11 · 9
HAR-RV + FHS	77.94	−4.0% [-5.7, -2.1]	<0.001	✓	5.13	11 · 11 · 8
HAR-RV + EVT	77.95	−4.0% [-5.7, -2.1]	<0.001	✓	5.12	11 · 11 · 7
GJR-GARCH-skewed-t	80.79	−0.5% [-2.0, +0.6]	0.222		4.89	9 · 10 · 10
GJR-GARCH + EVT (McNeil-Frey)	81.12	−0.1% [-0.2, -0.0]	0.063		5.25	11 · 10 · 10
CAViaR asymmetric slope	81.21	−0.0% [-1.2, +1.0]	0.788		4.95	8 · 8 · 9
GJR-GARCH + FHS (reference)	81.22	+0.0%	—		5.37	11 · 10 · 9
GARCH(1,1)-normal	81.57	+0.4% [-0.5, +1.4]	0.925		4.99	8 · 10 · 9
GARCH(1,1)-t	81.83	+0.7% [-0.0, +1.6]	0.969		5.37	8 · 10 · 7
RiskMetrics EWMA, normal	83.19	+2.4% [+0.8, +4.0]	0.999		5.48	10 · 10 · 3
Historical simulation (500d)	89.55	+10.3% [+3.4, +20.1]	0.996		5.17	11 · 5 · 1

Algorithm 4 Frozen forward test

1: **Freeze (once):** train Meridian M0 (5 seeds) and per-asset HAR on all data through 2026-10-29; save weights; record SHA-256 hashes in a manifest
2: **loop** each trading day after the close
3: fetch data into a separate snapshot; drop any bar of a trading day still in progress
4: verify every frozen file's hash; refuse to run on mismatch
5: for each new origin date: forecast with the frozen models; append to the log with a UTC timestamp
6: mark a row *pre-committed* only if its origin was the latest complete bar at logging time
7: **end loop**
8: **Score:** QLIKE of Meridian vs HAR on pre-committed core-asset rows whose outcome exists

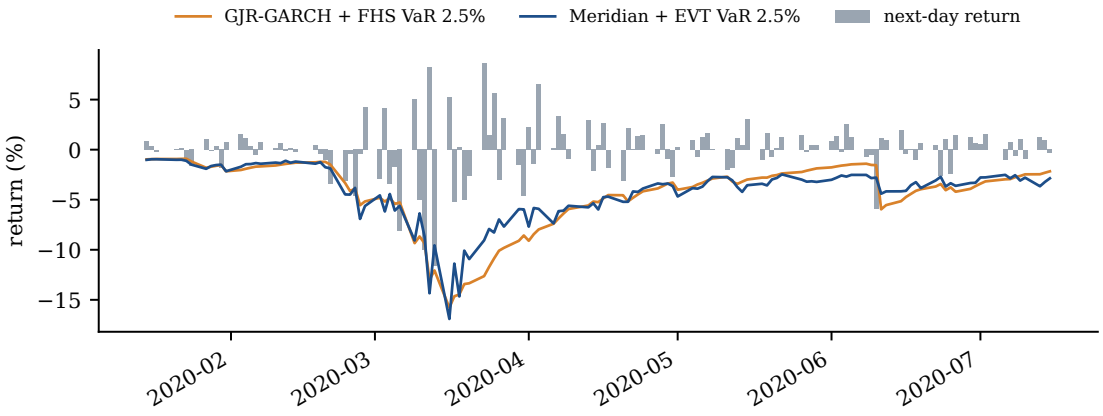


Figure 9: Illustration on real data: SPY next-day returns and 2.5% VaR forecasts, January–July 2020.

Table 23: Tail benchmark, development period (used only to choose the Meridian tail variant), 2.5%.

Model	FZ0	Δ vs reference [95% CI]	DM p	MCS	Viol. %	CC · DQ pass	ES ratio
HAR-RV + FHS	-3.7832	-0.0998 [-0.1247, -0.0724]	<0.001	✓	2.61	10/11 · 7/11	1.04
HAR-RV + EVT	-3.7830	-0.0981 [-0.1230, -0.0710]	<0.001	✓	2.46	9/11 · 8/11	1.04
Meridian + EVT	-3.7753	-0.0865 [-0.1159, -0.0585]	<0.001	✓	2.47	10/11 · 9/11	1.04
Meridian + FHS	-3.7746	-0.0862 [-0.1164, -0.0573]	<0.001	✓	2.63	10/11 · 8/11	1.03
Meridian + FHS + adaptive conformal	-3.7547	-0.0677 [-0.0976, -0.0393]	<0.001		2.53	10/11 · 9/11	1.04
LightGBM vol + FHS	-3.7514	-0.0595 [-0.0850, -0.0327]	<0.001		2.68	11/11 · 9/11	1.03
GJR-GARCH + EVT (McNeil-Frey)	-3.7034	-0.0002 [-0.0027, +0.0024]	0.452		2.23	9/11 · 9/11	1.05
GJR-GARCH + FHS (reference)	-3.7016	—	—		2.33	9/11 · 9/11	1.03
GJR-GARCH-skewed-t	-3.6911	+0.0161 [-0.0003, +0.0293]	0.983		1.88	5/11 · 8/11	1.05
GARCH(1,1)-t	-3.6842	+0.0204 [+0.0087, +0.0329]	0.999		2.21	8/11 · 10/11	1.06
GARCH(1,1)-normal	-3.6804	+0.0187 [+0.0036, +0.0379]	0.980		2.33	10/11 · 9/11	1.18
RiskMetrics EWMA, normal	-3.6207	+0.0676 [+0.0230, +0.1249]	0.995		3.26	5/11 · 3/11	1.21
Historical simulation (500d)	-3.5578	+0.1409 [+0.0936, +0.1979]	1.000		2.84	4/11 · 0/11	1.04

Table 24: Tail benchmark on the out-of-universe assets TLT and GLD, holdout, 2.5%.

Model	FZ0	Δ vs reference [95% CI]	DM p	MCS	Viol. %	CC · DQ pass	ES ratio
Meridian + EVT	-3.7059	-0.0638 [-0.1344, -0.0057]	0.015		2.71	2/2 · 2/2	1.00
Meridian + FHS	-3.7050	-0.0629 [-0.1333, -0.0050]	0.016		2.71	2/2 · 2/2	1.01
HAR-RV + EVT	-3.7023	-0.0602 [-0.1138, -0.0105]	0.007		2.82	2/2 · 2/2	1.02
HAR-RV + FHS	-3.6986	-0.0565 [-0.1097, -0.0079]	0.010		2.87	2/2 · 2/2	1.02
LightGBM vol + FHS	-3.6915	-0.0494 [-0.1081, +0.0021]	0.028		2.90	2/2 · 2/2	1.01
Meridian + FHS + adaptive conformal	-3.6870	-0.0449 [-0.1153, +0.0154]	0.071		2.58	2/2 · 2/2	1.01
GJR-GARCH-skewed-t	-3.6596	-0.0174 [-0.0393, +0.0025]	0.055		2.50	1/2 · 1/2	1.01
GARCH(1,1)-t	-3.6548	-0.0127 [-0.0256, +0.0010]	0.036		2.87	1/2 · 1/2	1.03
GARCH(1,1)-normal	-3.6453	-0.0032 [-0.0174, +0.0118]	0.332		3.03	1/2 · 1/2	1.11
GJR-GARCH + EVT (McNeil-Frey)	-3.6448	-0.0027 [-0.0086, +0.0025]	0.175		2.63	1/2 · 1/2	1.05
GJR-GARCH + FHS (reference)	-3.6421	—	—		3.08	1/2 · 1/2	1.02
RiskMetrics EWMA, normal	-3.6272	+0.0149 [-0.0111, +0.0458]	0.845		3.24	1/2 · 0/2	1.14
Historical simulation (500d)	-3.4763	+0.1658 [+0.0323, +0.3761]	0.981		3.43	1/2 · 0/2	1.05

Status. The models were frozen on 2026-11-01 16:19:03 UTC (code commit `108f349`, 54,481 training windows). 26 forecasts have been logged for 2 origin dates (2026-10-30 to 2026-11-02), all pre-committed (table 25).

Table 25: Forward-test log so far (13 assets per origin; 11 core assets are scored).

Forecast origin (close of)	Forecasts	Pre-committed	Logged at
2026-10-30	13	13	2026-11-01 16:19:03 UTC
2026-11-02	13	13	2026-11-03 08:44:03 UTC

First result. Scored on 2026-11-03 08:44:08 UTC: 1 day (origin 2026-10-30), 11 forecasts. Meridian QLIKE 0.3617 vs HAR 0.3201, i.e. $+13.0\%$ *worse* for Meridian. This is reported because it is real, but it carries essentially no information: on the holdout Meridian had the lower loss on only 61.2% of individual days, so a losing single day is expected about four times in ten.

How long until the forward test is informative? Using the holdout's mean daily loss difference and its HAC standard deviation, and assuming the holdout effect persists, a one-sided test at 5% needs about 197 trading days (≈9 months) for 50% power and about 451 trading days (≈1.8 years) for 80% power. Until then, the forward test is a monitoring tool, not evidence.

11 Discussion: what went well and what went wrong

What went well.

- Under a corrected, pre-declared protocol Meridian is the best of 15 volatility forecasters on an untouched seven-and-a-half-year holdout, with a consistent edge across all assets, almost every year, and all VIX regimes, largest in stress periods.
- The HAR-anchored design (start exactly at HAR, learn corrections) is what made the neural model competitive; the improvement over HAR survives against the strongest HAR form.
- The tail-risk application works: realized-volatility-filtered VaR/ES clearly beats the GARCH family and historical simulation, with well-calibrated ES.
- The project is now reproducible: forecasts are saved, every table is regenerated from them, and 11 tests guard the evaluation.

What went wrong.

- Earlier headline results were overstated by evaluation defects and had been published to GitHub (`ab041b0`); they are withdrawn.
- A data-quality problem (invalid FX bars) went unnoticed through several rounds of results.
- Three architectural improvement attempts (M1–M3) failed on the development period.
- Meridian's tail-risk advantage over a simple HAR-based tail is not significant.
- The regime, factor, network, combiner and portfolio modules have no validated numbers.
- Trading-strategy experiments recorded in the lab log did not reach the Sharpe target sought.

12 Targets, projections and timeline

Projection. The validated position is: a real, consistent improvement over HAR-RV of about the size of the pre-registered bar, best-in-class among the models tested on this data, and a tail-risk application that is

Table 26: Pre-registered and declared targets versus current validated status.

Target (source)	Current evidence	Status
$\geq 5\%$ QLIKE reduction vs HAR, DM $p < 0.05$ (§5; Am. 6 on holdout)	-7.5% [-11.3, -4.7], p <0.001; vs raw HAR -6.8% [-9.7, -4.5]	Met (point estimate)
Same bar with a confidence interval entirely beyond 5% (not pre-registered; stricter)	interval reaches -4.5%	Not met
Regime vs HMM (§5, Am. 2)	no audited result	Open
Tail VaR calibrated, Kupiec and Christoffersen not rejected (Am. 3)	Meridian + EVT: 11/11 and 11/11 at 1%; 11/11 and 11/11 at 5%	Met
Tail: lower FZ0 than GJR-GARCH + FHS, $p < 0.05$ (Am. 7)	-0.1218, p <0.001	Met
Forward confirmation of the volatility gain (Am. 7)	1 day scored	Pending

strong against classical models but not distinguishable from a HAR-based tail. We do *not* project trading profitability. The time estimates in table 27 are estimates, not measurements, except where the basis says otherwise.

Table 27: Remaining work and estimated time.

#	Work item	Estimated time	Basis
1	Run the forward test daily	≈9 months (50% power); ≈1.8 years (80% power)	power calculation from holdout statistics (section 10)
2	Re-validate regime, factor, network, combiner, continual-learning and portfolio modules	about 1–2 working days per module (estimate)	the tail module took one working session; its full benchmark computes in 99 s
3	Full 15-model comparison at 5 and 22 days	about 1 working day (estimate)	the volatility benchmark runs in 18 min
4	Intraday (5-minute) data: HARQ, HAR-CJ, Realized GARCH, cleaner target	cannot be estimated	depends on data access and cost
5	More assets (bonds, commodities, futures) to tighten intervals	about 1–2 working days plus compute (estimate)	pipeline already supports extra assets
6	Push `harness-audit` and replace the withdrawn claims on GitHub	under 1 hour	ready; awaiting decision

13 Limitations

- **Variance proxy.** Daily range-based realized variance only; no intraday data, so HARQ, HAR-CJ, Realized GARCH and true semivariances are untested and the SHAR model is an approximation.
- **Universe.** 11 core assets, free and delayed data; results may not transfer to other asset classes, though the TLT/GLD check is encouraging.
- **Statistics.** The DM test omits the Harvey–Leybourne–Newbold small-sample correction [22]; with

2,168 holdout dates its effect is negligible. Several comparisons are reported; the primary one was declared in advance.

- **Economic value.** A better volatility or VaR forecast is not a trading strategy; no Sharpe ratio, return or P&L is claimed.
- **Other modules.** The repository contains 68 scripts, including regime, factor, network and portfolio experiments; only the volatility and tail-risk results in this report are validated.

14 Conclusion

After correcting eight evaluation defects and re-running every benchmark under a pre-declared development/holdout protocol, Meridian remains the best of the volatility forecasters tested, improving on per-asset HAR-RV by -7.5% (-6.8% against the strongest HAR form) with consistent gains across assets, years and market regimes. The improvement meets the pre-registered bar on its point estimate, but its size is uncertain at the 5% threshold. As a tail-risk engine it clearly beats classical models but not a HAR-based tail. The decisive remaining evidence is the forward test, which has just begun and will need on the order of 451 trading days to be conclusive.

A Reproducibility

Table 28: Commands that regenerate every result in this report.

Command	Produces
`python -m pytest tests -q`	11 correctness tests
`BENCH_VARIANTS=M0 BENCH_ENSEMBLE=5 python scripts/benchmark_vol.py`	volatility forecasts (18 min, commit `ab041b0`)
`python scripts/score_vol_benchmark.py`	`results/benchmark_vol.json`
`python scripts/benchmark_tail.py`	tail forecasts (99 s)
`python scripts/score_tail_benchmark.py`	`results/benchmark_tail.json`
`python scripts/forward_test.py predict | score`	forward log and score
`python scripts/build_report_tex.py; latexmk -pdf main.tex`	this report

Table 29: Automated correctness tests (all passing).

File	Test
`test_harness.py`	`test_qlike_nonnegative_and_minimized_at_truth`
`test_harness.py`	`test_calibration_has_no_lookahead`
`test_harness.py`	`test_hac_dm_is_calibrated_under_null`
`test_harness.py`	`test_score_models_equal_models_not_significant`
`test_harness.py`	`test_no_train_label_crosses_test_start`
`test_harness.py`	`test_no_degenerate_rv_rows_after_cleaning`
`test_tail.py`	`test_fz0_minimized_at_true_var_es`
`test_tail.py`	`test_fz0_differences_scale_invariant`
`test_tail.py`	`test_tick_minimized_at_quantile`
`test_tail.py`	`test_backtests_calibrated_under_null`
`test_tail.py`	`test_filtered_tail_has_no_lookahead`

B Glossary

QLIKE Loss for variance forecasts, $\mathrm{RV}/\hat{\sigma}^2 - \ln(\mathrm{RV}/\hat{\sigma}^2) - 1$; robust to noise in the RV proxy.

Table 30: Software environment of the audited runs.

Component	Version
Python	3.12.13
PyTorch	2.13.0
arch	8.0.0
scikit-learn	1.9.0
LightGBM	4.7.0
SciPy	1.18.0
pandas	3.0.5
NumPy	2.5.1
Platform	macOS-26.5-arm64-arm-64bit

Table 31: Code inventory.

Folder	Python files	Lines of code
`meridian/`	17	2,566
`scripts/`	68	9,313
`tests/`	2	153

Table 32: Repository history (branch `harness-audit`; `main` = the commit pushed to GitHub on 2026-09-08).

Commit	Date	Message
0a6cfcc	2026-11-03	Add LaTeX project report generator; forward test ignores in-progress bars
d6c444b	2026-11-01	Add generated PDF report of the audited volatility and tail-risk benchmarks
f3932c5	2026-11-01	Add audited tail-risk (VaR/ES) benchmark vs 13 risk models and a frozen forward test
108f349	2026-11-01	Withdraw pre-audit numbers from ARCHITECTURE, WORLD_MODEL and RESEARCH docs
df4e93b	2026-11-01	Audit the volatility benchmark harness; withdraw invalid numbers; re-benchmark vs 15 models
ab041b0	2026-09-08	Meridian: HAR-anchored, calibrated volatility core beats HAR-RV (-8.7% QLIKE, p<0.001)

HAR-RV Regression of tomorrow's log RV on daily, weekly and monthly average RV.

DM test Diebold–Mariano test of equal predictive accuracy.

MCS Model Confidence Set: the set of models not significantly worse than the best.

VaR / ES Value-at-Risk (loss quantile) and Expected Shortfall (mean loss beyond VaR).

FHS / EVT Filtered historical simulation; extreme-value (Generalized Pareto) tail.

FZ0 Joint scoring function for (VaR, ES).

Holdout Data never used for any modelling choice (2019-01-01 to 2026-06-29).

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